

**Enhancing Emergency Preparedness through Immersive Virtual Reality: A Comparative
Study of Stress Responses in Radiological Simulations**

by

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The student author, whose presentation of the scholarship herein was approved by the program of study committee, is solely responsible for the content of this thesis. The Graduate College will ensure this thesis is globally accessible and will not permit alterations after a degree is conferred.

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NOMENCLATURE

IVR – Immersive Virtual Reality

RSSQ - Radiation Source Stress Questionnaire

RW – Real world

VR - Virtual Reality

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ABSTRACT

Immersive virtual reality (IVR) is increasingly used in hazardous materials training, but questions remain about how accurately it replicates the psychological and behavioral stress responses experienced in real-world (RW) scenarios. This study investigates whether an IVR-based simulation of a radiological exposure event can evoke radiation-induced stress levels comparable to those reported in an equivalent RW simulation among experienced first responders. Drawing upon literature that highlights the complex hazards faced by emergency personnel, this study emphasizes the distinct challenges associated with radiation exposure—an invisible yet highly consequential threat. The research was motivated by a recognized gap in current training paradigms, particularly the limitations of traditional methods to realistically replicate the psychological intensity and situational complexity of radiological emergencies.

A between-subject experimental design was employed with 55 professional firefighters from central Iowa, who were randomly assigned to participate in either an IVR or a RW simulation of a radiological incident. The methodology included pre- and post-simulation surveys, a scenario-based response task, and a custom-developed Radiation Source Stress Questionnaire (RSSQ) to assess stress related to proximity to a radiation source. Confirmatory factor analysis of the RSSQ revealed two latent constructs: (1) Behavioral and Cognitive Response to Presence of Radiation Source, and (2) Perceived Risk and Emotional Concern.

Statistical comparisons indicated that while behavioral and cognitive responses were similar across both groups, participants in the IVR condition reported significantly higher levels of emotional concern and perceived risk. Multiple linear regression analysis showed that risk-taking propensity was a significant negative predictor of stress responses, and perceived life stress had a marginal positive effect. Additionally, task completion times were longer for the

IVR group, potentially leading to an enhanced cognitive load during virtual simulation; however, given the short duration of the task (4–5 minutes), it is reasonable to assume that time-induced cognitive load was not meaningfully elevated in either group.

These findings suggest that IVR training can simulate certain stress responses to a degree equal to—or exceeding—those experienced in real-world conditions. This divergence presents both opportunities and cautions: elevated emotional arousal in IVR may enhance memory retention, but it could also distort perceived risk if not properly moderated.

Overall, this study contributes to the growing discourse on the role of immersive technologies in emergency responder training and recommends further longitudinal research incorporating physiological metrics to better understand the implications of simulated stress exposure in high-risk environments.

CHAPTER 1. GENERAL INTRODUCTION

Hazard training is essential for preparing first responders to handle dangerous situations effectively, often serving as the crucial factor between life and death. As Baetzner et al. (2022) note, the efficacy of a responder's actions in hazardous scenarios is heavily dependent on the training methods and their quality. Traditionally, hazard training has been delivered through lectures, discussions, and RW scenario exercises. However, technological advancements have introduced alternative methods such as computer-based training, video instruction, and IVR, all of which offer new avenues for enhancing learning outcomes.

The importance of effective training for responding to hazardous situations is underscored by the observation that many health professionals and first responders lack sufficient preparation to respond optimally in chemical, radiological, biological, and nuclear (CRBN) incidents (Kako et al., 2018). Moreover, the components that define effective hazard training have evolved over the years, necessitating regular updates to ensure that training methods remain current and effective (Rosen et al., 2023). Research consistently demonstrates that training significantly enhances the ability to perform complex tasks, with higher-fidelity simulations—those that closely mimic RW conditions—proving particularly effective in improving performance under pressure (Burke et al., 2006; Elendu et al., 2024).

Recent advancements in IVR technology have shown promising potential for creating realistic training scenarios. For instance, the IVR experience “Richie’s Plank Experience” has been effective in eliciting genuine biophysical responses from users, with individuals who experience acrophobia displaying measurable signs of fear when immersed in a virtual high-altitude environment. Despite such promising results, there remains a notable gap in research regarding the immersive qualities of IVR—specifically, how the sense of presence and

immersion contribute to realistic responses in the context of radiation hazards. Radiation presents a unique challenge for simulation training due to its invisible nature and often delayed physical effects.

The following section explores the current state of research on IVR-based hazard training, with a particular focus on radiation hazards. The objective of this research is to evaluate the potential of IVR technology in replicating realistic stress responses, thereby contributing to the enhancement of training protocols for radiation emergencies.

CHAPTER 2. LITERATURE REVIEW

What are the Hazards

Hazards encountered by first responders are multifaceted and complex. First responders—including firefighters, paramedics, police officers, and other public safety personnel—regularly face life-threatening situations that range from fires and hazardous materials incidents to violent events (International Atomic Energy Agency, 2006). Upon arrival at an incident scene, these professionals must assess their surroundings rapidly and make critical decisions to protect both the affected public and themselves. However, the mere accumulation of experience, such as that of veteran firefighters, does not necessarily translate to an enhanced ability to recognize and respond to hazards (Bayouth et al., 2013; Bayouth & Keren, 2019; Hardison & Gray, 2021; Keren et al., 2013a,b). This highlights the essential role of targeted training programs designed to sharpen hazard recognition skills across all levels of experience.

In general, first responders confront a broad spectrum of hazards that extend beyond immediate physical dangers. They are often the first on the scene of accidents and disasters, environments that are frequently unpredictable and fraught with unknown risks (Turner et al., 2016). These unpredictable conditions contribute to a higher incidence of occupational stress injuries, further complicating the challenges inherent in their work (Antony et al., 2020). Physically, responders are susceptible to a range of injuries—from minor cuts and bruises to more severe outcomes such as heat stress, fractures, and even fatalities (Khatri et al., 2019). In addition to these physical risks, first responders also experience significant mental health challenges, including elevated rates of depression, post-traumatic stress disorder (PTSD), and suicide (Horan et al., 2021).

To mitigate these risks, first responders rely on a combination of personal protective equipment (PPE) and comprehensive training that prepares them for a variety of scenarios. This training not only focuses on the physical aspects of hazard mitigation but also attempts to address mental health concerns. Recent efforts to tailor mental health resources to the specific needs of first responders have shown promise in reducing the incidence and severity of these issues (Medcalf et al., 2021; Wright et al., 2022).

Radiation hazards present a unique set of challenges compared to other types of threats. Unlike visible hazards, radiation is unseen and cannot be detected without specialized equipment, which must then be correctly interpreted by the responder. The relative rarity of radiation incidents often results in fewer resources being allocated to radiation-specific training, leaving many local responders with limited real-world experience in managing such scenarios (International Atomic Energy Agency, 2006). Despite the low probability of a radiation incident, the potential consequences of radiation exposure can be high; resulting in the potential for health impacts such as cancer and acute radiation syndrome, as well as public and emergency worker panic (Al-Ibraheem et al., 2024).

To minimize the risks associated with radiation exposure, safety protocols adhere to the ALARA principle— “as low as reasonably achievable” (Hendee & Edwards, 1986). This principle is operationalized through three key strategies: 1) reducing the time spent near the radiation source, 2) maximizing the distance from it, and 3) utilizing appropriate shielding to block or reduce radiation levels. Together, these measures form the cornerstone of radiation safety protocols, aiming to protect first responders from the potential dangers of radiation while they carry out their essential duties.

For the purposes of this research, radiation is the hazard of interest due to the potential risk of harm to first responders by compromising their physical, mental, or operational integrity if not properly controlled. This description integrates both the immediate dangers, and the latent risks associated with radiation exposure.

Traditional training methods, though valuable, often lack the specificity and realism required for effective radiation emergency preparedness. IVR has emerged as a promising tool that can simulate radiological incidents in a controlled yet realistic manner. IVR-based training offers the potential to replicate the complex conditions of radiation emergencies without incurring actual radiation exposure, enabling first responders to safely practice critical decision-making and stress management in scenarios that closely mirror real-life challenges (Nomura et al., 2024; Rebmann et al., 2019). By integrating IVR simulations with established training protocols, the effectiveness of preparing emergency responders to face the multifaceted dangers of radiological events can be enhanced. In essence, if IVR can deliver a training level comparable to that of a RW simulation, the efforts required for conducting radiation-related emergency response training could be significantly less costly and eliminate the need for radiation exposures from training sessions, while also having the potential to enhance the level of training due to the various affordances facilitated by IVR technology.

Gaps in First Responders' Understanding of Radiation Hazards

First responders may face critical challenges during radiological incidents, yet many lack a proper familiarity with the fundamental aspects of radiation. As Leek et al. (2024) indicate, responders frequently misjudge the relationship between perceived risk and actual risk. This discrepancy is largely attributable to the rarity of radiological events, which limits opportunities

for gaining practical experience. Consequently, many responders are ill-prepared to accurately assess and manage the unique hazards posed by radiation.

This evident knowledge gap emphasizes the urgent need for robust, specialized training in radiological emergencies. However, implementing such training programs is not without its challenges. Xu et al. (2024) note that many jurisdictions struggle with budgetary constraints, making it difficult to allocate the necessary resources for comprehensive radiological training. As a result, numerous firefighters and first responders have received little to no instruction on how to effectively respond to radiological incidents (Rebmann et al., 2019).

Despite these challenges, evidence suggests that targeted training can significantly enhance the readiness and confidence of emergency personnel. For example, Reilly et al. (2007) demonstrated that training in response to chemical, biological, or radiological events substantially increases the comfort level of emergency medical services providers. This improvement in preparedness fosters better decision-making in the field and helps mitigate the risks associated with these complex hazards.

Hazard Training: Stress, Cognitive Appraisal, and Skill Transfer

Stress exerts a profound influence on decision-making processes, often shifting individuals from deliberate, analytical reasoning toward more habitual, automatic responses (Arnsten et al., 2017). This phenomenon reinforces the importance of understanding the physiological underpinnings of stress and the cognitive processes that mediate its effects.

The Cognitive Appraisal Theory offers a robust framework for this understanding. It posits that stress arises from an individual's evaluation of a situation exceeding their available coping resources (Lazarus & Folkman, 1984). In this view, stress is not merely a reaction to

external events; rather, it is shaped by the subjective interpretation of those events. As Cohen, Janicki-Deverts, and Miller (2007) argue, these personal evaluations are critical in determining the overall impact of stress on health and wellbeing.

This cognitive perspective is particularly pertinent in the context of radiation emergency response. First responders, when confronting CBRN incidents experience stress due to anticipatory anxiety, which can have disastrous effects on the first responder's ability to perform (Giaume et al., 2024). This stress is caused by the perceived danger of the CBRN incident (Lazarus and Folkman, 1984). The Cognitive Appraisal model suggests that even when exposed to identical external conditions, individual differences in perception and confidence can result in widely varying stress responses.

Adding another layer of complexity is the concept of secondary traumatic stress, the emotional duress that arises when individuals are indirectly exposed to trauma, such as through hearing about the distressing experiences of others (Naturale, 2007). In high-risk settings like radiological emergencies, initial stress reactions can be compounded by subsequent exposures to traumatic outcomes. This reality highlights the necessity for structured interventions, such as Critical Incident Stress Debriefing (CISD), which are designed to address both the cognitive and emotional components of stress and thereby enhance coping and resilience (Mitchell, 2008).

Given the multifactorial nature of stress, the Cognitive Appraisal framework provides a comprehensive lens through which to understand and manage the challenges faced by radiation responders. This approach accounts for immediate biological responses and emphasizes the subjective perception of threat and the efficacy of personal coping strategies (Lazarus & Folkman, 1984; Sapolsky, 2004). Recognizing this duality is essential for the development of effective training programs and support mechanisms—both in IVR simulations and RW

scenarios—that prepare responders to manage the complex stressors associated with radiological incidents.

RW training often incorporates stress induction to better prepare individuals for the pressures of actual emergencies. For example, sports training frequently includes stress-inducing drills to simulate the high-pressure conditions of competitive events (Low et al., 2022), and military training programs are explicitly designed to improve stress resilience (Jensen et al., 2020). Additionally, firefighter training exercises that simulate live fire conditions have been found to elicit stress levels similar to those experienced during real emergency responses (Gonzalez et al., 2023).

Similarly, IVR-based training scenarios can induce significant psychological stress, thereby replicating many of the emotional and physiological responses characteristic of RW emergencies. In one study, a computerized simulation of a fire scene effectively elicited psychological stress among participants (Throne et al., 2000). Further investigations into IVR-induced fear have highlighted that the plausibility illusion—the perception that the simulated event is realistic—plays a more pivotal role in eliciting fear than the visual realism of the environment itself (Lin, 2017). Consistent findings indicate that IVR scenarios can induce not only immediate fear but also anxiety and varying levels of stress, with some individuals experiencing residual effects even into the following day (Lin, 2017; Yeh et al., 2018; Finseth et al., 2023).

Comparative Analysis

Over the past two decades, there has been a growing interest in developing IVR applications to enhance training across industries, governmental agencies, and healthcare. There has been interest in using IVR as a means for enhancing training and learning for decades.

Research by Rose et al. (2000) demonstrated that a virtual environment can achieve levels of learning transfer comparable to traditional RW training, suggesting IVR's viability as an alternative approach. Building on this, Oren et al. (2012) found that in a simple puzzle assembly task, participants spent more time engaged in VR training and outperformed those trained in a RW setting. In the healthcare sector, for example, VR-based training has been extensively studied. Kyaw et al. (2019) conducted a meta-analysis which found that VR training significantly improved knowledge and skills outcomes compared to other digital education platforms, although the authors stated that further research is needed to assess IVR's impact on attitudes, satisfaction, cost-effectiveness, and behavior change.

Similarly, studies focusing on nursing education by Chen, Leng, et al. (2020) and Chen, Huang, et al. (2019) confirmed the positive impact of VR on knowledge acquisition, though they reported mixed results regarding skill acquisition, performance efficiency, and task completion times. These studies also observed that, relative to control groups, VR training did not consistently enhance satisfaction or confidence among trainees. Supporting these findings, Zhao et al. (2020) conducted a meta-analysis on the effectiveness of VR in teaching anatomy to paramedics, further underlining VR's role in knowledge transfer despite variable impacts on other performance metrics.

In another study, Mills et al. (2019) compared VR-based training to live simulations for paramedics. While no significant differences in outcomes or satisfaction levels were observed, VR training emerged as highly cost-effective—costing less than 10% of the expenses associated with live simulations. Beyond healthcare, emergency response training has also attracted considerable attention, with multiple studies (e.g., Bayouth & Keren, 2019; Bayouth et al., 2013;

Engelbrecht et al., 2019; Finseth et al., 2018, 2022, 2023, 2025; Keren et al., 2013a,b) exploring VR's utility in high-stress, dynamic scenarios.

Comparative research between VR and traditional RW training further highlights VR's potential. Cooper et al. (2021) reported that both versions of IVR training—with and without multisensory cues—facilitated faster task completion than traditional methods. In the realm of firefighter training, Narciso et al. (2023) found a nuanced balance between the training modalities: while RW training induced higher stress levels, VR training yielded a greater degree of knowledge transfer. The authors concluded that, although VR was marginally less effective in some aspects, its benefits in retaining knowledge make it a valuable supplementary tool.

Emotional responses also vary between VR and RW training. Kleygrewe et al. (2023) investigated VR scenario-based training for police officers and found that, although RW training led to higher heart rates and greater physical activity, the mental effort required in VR was significantly higher. Their study concluded that VR could elicit perceived stress, mental effort, and physiological responses that are comparable to those observed in RW training scenarios.

As presented above, VR has emerged as a promising tool for response to hazardous conditions, offering immersive and dynamic simulations that traditional methods may struggle to replicate. VR training environments have demonstrated the capacity to evoke emotional responses that are beneficial to the learning process. Specifically, research has shown that VR can increase positive emotions while reducing negative ones, aligning with the control value theory of achievement emotions (Nomura et al., 2024). Moreover, studies have indicated that the skills acquired through VR-based training are transferred at a level comparable to those gained from real-world simulations (Rose et al., 2000). One of the significant advantages of VR is its ability to simulate dangerous scenarios that would be too risky to recreate in live training,

thereby allowing trainees to safely experience and learn to manage extreme hazards (Holuša et al., 2023).

VR Challenges

While VR offers numerous advantages for training for handling hazard, several challenges must be addressed to ensure its effective implementation. One major concern is the cost factor. Although the initial investment in VR hardware, software, and infrastructure can be considerably higher than traditional RW simulations, VR has the potential to be more cost-effective over time. For instance, Farra et al. (2019) found that despite higher upfront expenses, a VR-based training setup can become more economical after approximately three years, largely due to its scalability and time flexibility. However, Pedram et al. (2021) highlights those ongoing costs, such as those for headsets, maintenance, and periodic upgrades, can add up. Additionally, there are costs associated with the learning curve and integration of VR technology into existing training frameworks. Other practical challenges include the inability of VR to simulate certain sensory inputs like heat and smell, the occurrence of simulator sickness in some users, limitations in adaptive user interaction, and a degree of reluctance among some individuals to adopt VR as a training tool.

Another significant challenge is the learning curve associated with VR training. While VR enables safe repetition, which is essential for mastering complex tasks, its effectiveness is contingent upon sustained motivation among trainees. Fartoussi et al. (2023) note that if motivation diminishes, the benefits of repeated practice may not be fully realized. Although VR has been shown to generate higher levels of engagement and interest compared to traditional RW practices (Markwell et al., 2023), this enthusiasm may be primarily driven by the novelty of the technology rather than its long-term educational value, as suggested by Adachi et al. (2017).

Consequently, addressing these challenges is critical for ensuring that VR not only enhances training outcomes in the short term but also serves as a sustainable, effective tool in the long run.

IVR is a multifaceted concept that plays a critical role in the overall effectiveness of VR training. According to Slater (2018), immersion is defined as an objective property of a VR system—its ability to support natural sensorimotor contingencies for perception. In contrast, presence refers to the subjective psychological sense of "being there" within the virtual environment. Slater posits that higher levels of immersion generally correlate with a stronger sense of presence, which can enhance training experience by making virtual scenarios feel more realistic.

Several technical factors contribute to a more immersive VR experience. Improvements in display resolution, field of view, stereoscopy, head tracking, full-body motion capture, as well as audio and haptic feedback, all play pivotal roles in deepening the immersive quality of a VR system. It is important to recognize, however, that immersion is not equivalent to belief; even when users are aware that they are interacting with a simulation, their physiological and emotional responses to virtual stimuli can be as genuine as those experienced in RW situations.

The benefits of high immersion levels in VR extend beyond just creating a realistic environment. Research by Lønne et al. (2023) indicates that VR environments tend to be more immersive and yield greater positive effects compared to traditional desktop-based systems. Furthermore, immersion in VR has been associated with numerous cognitive and motivational benefits. As noted by Guerra-Tamez (2023), enhanced immersion can boost motivation and curiosity, promote reflective thinking, and improve overall cognitive performance. These benefits are particularly evident in educational settings, where immersive VR experiences can increase the perceived value of learning and foster deeper engagement with the material.

Overall, VR-based simulation for training for handling hazards offers an innovative, immersive, and engaging learning experience with clear cognitive and motivational benefits, its full potential depends on overcoming challenges. Balancing the high upfront and ongoing costs, addressing sensory and technical limitations, and ensuring sustained trainee motivation is crucial for long-term success. The success of VR based hazard trainings depends on the ability of the simulation to address these concerns while still maintaining its benefits.

Overarching Research Question

As the use of VR continues to expand in emergency responder training, there is growing interest in understanding how immersive simulations compare to RW scenarios—particularly when it comes to psychological and behavioral responses. While VR offers unique advantages such as safety, repeatability, and cost-effectiveness, it is not yet clear whether it can reliably replicate the sense of stress and concern that responders experience when faced with actual radiological hazards. Several studies have explored risk perception and realism in simulations, but the question remains: does IVR trigger stress responses that are meaningfully similar to those evoked by RW conditions?

This study aims to evaluate this gap by comparing radiation-induced stress between responders who participated in an IVR-based simulation and those who took part in a RW version of the same exercise.

The overarching research question guiding this research is **To what extent does an immersive virtual reality simulation of a radiological scenario produce stress responses that are comparable to those experienced in a real-world simulation?**

This study, guided by the research question above, seeks to inform if IVR can be used more effectively in radiation related hazardous materials training and to determine whether it can serve as a reliable proxy for real-world exposure in preparing responders for high-stakes environments.

CHAPTER 3. METHODOLOGY

To explore the research question herein, a between-subject experimental design was employed. Participants were divided into two groups as follows: 29 subjects (53%) engaged in the IVR simulation of a radiation incident, and 26 (47%) participated in a RW simulation replicating similar conditions. This design enables a direct comparison of stress responses across the IVR and RW environment. Description of the scenarios are provided below. Figure 1 presents an overview of the experimental process.

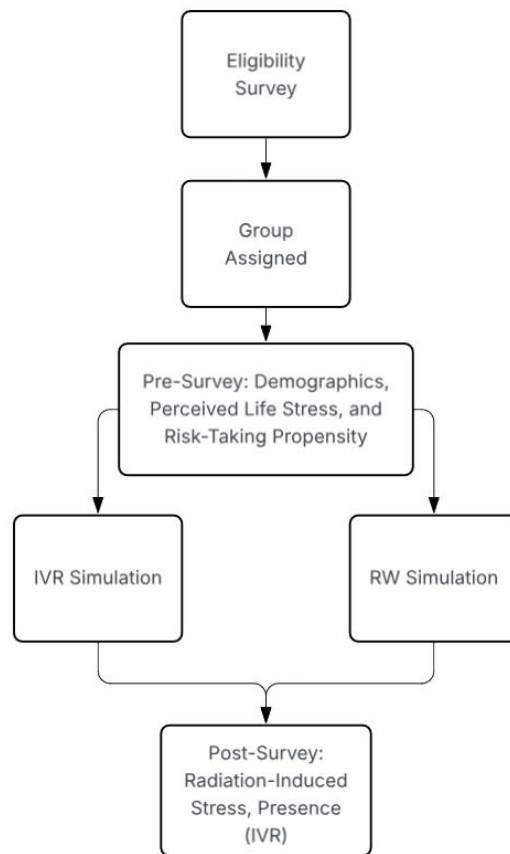


Figure 1: Overview of the experimental process

Subjects and Condition Assignment

The participants in this study consisted of career firefighters from central Iowa, all of whom have professional experience responding to emergency situations, including hazardous material incidents. Firefighters were selected due to their direct involvement in radiation emergency response scenarios and their familiarity with high-stress environments. The inclusion of experienced responders ensures that the findings are applicable to real-world emergency response training and preparedness.

To ensure appropriate placement within the experimental conditions, all participants underwent an initial eligibility assessment through a brief survey. This survey was designed to determine whether individuals met the criteria for participation in the study and whether they were eligible for the IVR simulation. The assessment included questions related to prior experience with VR, susceptibility to motion sickness, and any medical conditions that could interfere with participation in either the IVR or RW simulation condition.

Following the eligibility assessment, participants were assigned to one of two conditions:

- IVR simulation
- RW simulation

Participants were assigned to experimental conditions randomly. Although, if a participant indicated a higher susceptibility to simulation sickness during the eligibility assessment, they were placed in the RW simulation condition to mitigate potential adverse effects. Any individuals deemed ineligible for both conditions were excluded from the study.

Each participant was assigned an anonymous reference identification number. This ID was used to securely link their responses from the pre-survey, post-survey, and observational

data, maintaining confidentiality while allowing for comprehensive data analysis. Each participant reviewed a consent form tailored to their assigned condition, outlining the specific risks and expectations associated with their participation. The consent process ensured that participants were fully informed about the nature of the study, the potential risks involved, and their right to withdraw at any time.

The study was approved by the Iowa State University Institutional Review Board (IRB ID 24-479) to ensure compliance with research protocols and ethics.

The following are the descriptions of each condition's setting.

Real World Condition

Participants assigned to the RW condition performed the radiation incident response simulation in an open space measuring at least 25 x 25 feet. At the far end of the space, five boxes with various labels, including radiation-related hazard labels, were positioned. The labels on the boxes were designed to mimic those that could be found in actual hazardous material settings. Figure 2 provides an example of the radiation-related labels on one of the boxes, while Figure 3 illustrates the overall setup, including the placement of the five boxes within the simulation environment.



Figure 2: Example of radiation-related labels on a box.



Figure 3: RL condition setup with five labeled boxes

A key aspect of the RW condition was ensuring that participants believed that the training environment incorporated the use of real radiation sources. The research team was careful to portray the RW condition as if real radiation sources were present, and participants were not told that the radiation readings were in fact produced by a simulation device until after they completed the entire study.

To initiate the exercise, a research team member read a standardized script describing the simulated radiation incident scenario as follows:

“During this training exercise, you will approach a scene containing several boxes labeled as radioactive materials. Your goal is to identify which box contains the damaged radiation source or radiation source that is outside its shielding and report its isotope and transport index. A Radiation Safety Expert will monitor your exposure and ensure your safety throughout the process.”

Participants were then instructed to assess the environment and identify the box with the highest radiation level. Upon identifying the correct box, they were required to read and interpret information from multiple labels affixed to it. The main goal of this exercise was to ensure the participant remain in the radiation exposure environment for several minutes.

To conduct their assessment, participants were equipped with a Teletrix TXR-55 Radiation Monitor Simulator while a researcher was equipped with a Teletrix TXT-55 Digital Remote Controller. The TXR-55 Radiation Monitor Simulator was used to measure simulated radiation levels, while the TXT-55 Digital Remote Controller allowed for interactive engagement with the training environment. Figure 4 provides an image of the TXR-55 Radiation Monitor Simulator, and Figure 5 displays the TXT-55 Digital Remote Controller used during the exercise.



Figure 4: Teletrix TXR-55 Radiation Monitor Simulator.



Figure 5: Teletrix TXT-55 Digital Remote Controller.

During the simulation, a member of the research team acted as a Radiation Safety Expert (RSE) to add confidence to the sense of a realistic radiation simulation event. The researcher playing the role of RSE pretended to open the radiation source inside the designated box while actively manipulating radiation readings using the Teletrix TXT-55 Digital Remote Controller. This allowed for dynamic changes in radiation levels as perceived by the participants, simulating real-time exposure conditions.

The researchers adjusted the TXR-55 Radiation Monitor Simulator readings based on their estimation of the participants' distance from the simulated source to proximate RW conditions. The radiation level on contact with the 'damaged box' was adjusted to 26 R/h to simulate potential heightened exposure similar to a compromised containment scenario. The researchers would adjust the level of radiation based on an estimated inverse square law principle as the participant increased their distance from the damaged box.

To extend participant engagement in the simulated radiation scene and enhance the psychological sensation of exposure, additional tasks were introduced. After identifying the box with the highest radiation level, participants were required to examine and document other hazard labels on the box. Additionally, they measured and recorded radiation readings from multiple points on the radiation box, including the top, end, and one side. These tasks prolonged the interaction with the simulated hazardous environment, aiming at ensuring they had sufficient time within the environment to attribute their level of stress on the post simulation survey.

At the conclusion of each experimental session, when the participants concluded with all experimental surveys and activities, the researcher playing the RSE role conducted a structured debriefing with the participants. During this session, the researchers disclosed that no actual radiation source was present during the exercise and the reasoning for not sharing this during the

study activities. Participants were informed that radiation simulation equipment—specifically, the Teletrix TXR-55 Radiation Monitor Simulator and the Teletrix TXT-55 Digital Remote Controller—was utilized to create an authentic sense of a radiation field. This clarification ensured that participants understood the simulated nature of the exercise.

IVR Condition

The IVR condition was designed to closely replicate the tasks performed in the RW condition while accommodating the differences inherent to the virtual environment. Participants used an Oculus Quest 3 VR headset to engage with the radiation response scenario (Figure 6), allowing them to navigate and interact with the virtual environment. The simulation was conducted indoors in a space large enough to use the VR headset.



Figure 6: Screenshot of the IVR radiation simulation environment.

Before beginning the main simulation, participants completed an IVR tutorial designed to familiarize them with the IVR-specific controls and interactions necessary for the exercise. This tutorial ensured that all participants could effectively navigate the virtual environment and operate the radiation monitoring tools.

Unlike the RW condition, where the researcher acting as the RSE dynamically controlled the radiation readings, the radiation levels in IVR were algorithmically determined based on the inverse square law. Radiation exposure readings were calculated as a function of the participant's distance from the surface of the damaged virtual box.



Figure 7: Screenshot of the radiation simulation monitor in the VR.

Participants were asked to identify the highest radiation level and examine radiation hazard labels on the virtual boxes, mirroring the tasks performed in the RW condition.

Data Collection

Pre-Experiment Surveys

Before beginning their assigned condition (IVR or RW), participants completed a pre-simulation survey administered through Qualtrics. The pre-survey served three primary objectives:

1. Eligibility Screening

- Participants were screened for general eligibility based on professional background, training, and medical exclusions (See Appendix A). Questions assessed whether participants were professional emergency responders, between 18 and 65 years old, and had prior training on the use of radiation detection equipment.
- Additional exclusion criteria were applied, such as pregnancy, medical conditions (e.g., pacemakers, seizure disorders, or psychiatric conditions), and sensitivity to motion sickness (for IVR participants). Those who did not meet eligibility criteria were excluded from the study.

2. Background and Demographics

- Participants provided demographic information such as age, gender, ethnicity, marital status, education level, and number of dependents (See Appendix B).
- Additional questions gathered information on radiation-related experience, training certifications (e.g., HAZMAT certification level), frequency of radiation equipment use, perceived risk of radiation incidents, and confidence in responding to such incidents.

3. Pre-Experiment Psychological and Behavioral Measures

- Participants completed two validated self-report questionnaires to assess individual differences that could influence their responses to the radiation incident simulation:
 - Risk-Taking Propensity Questionnaire (Zhang et al., 2018): Evaluated participants' general attitudes toward risk, including their likelihood of engaging in risky behaviors and comfort with risk.
 - Perceived Stress Questionnaire (Cohen et al., 1983): Measured participants' pre-experiment stress levels over the past month, including how often they felt overwhelmed, in control, or emotionally stressed.

Experimental Data Collection

During the experimental session, the only quantitative behavioral metric recorded was task completion time.

Post-Experiment Questionnaires

After completing the simulation, participants filled out several questionnaires, administered through Qualtrics, to assess their experience, perceived radiation induced stress levels, and Presence in IVR (IVR condition only). Another survey assessing hypothetical communication with a superior officer was administered but is not in the scope of this study.

1. Presence in IVR Questionnaire (Witmer & Singer, 1998) (IVR condition only)

- Evaluated participants' sense of immersion, interaction quality, and sensory engagement in the IVR environment. This measure was used to determine how effectively IVR replicated a RW emergency scenario.

2. Radiation Source Stress Questionnaire (RSSQ)

- Assessed participants' psychological stress response to the simulated radiation threat. It measured awareness of the radiation source, perceived danger, concern about exposure, comfort near the source, hesitation, and impact on focus. The development of this questionnaire was founded on Slovic's Perception of Risk (2016) work on risk perception, which emphasizes that individuals evaluate risks based on subjective factors such as dread, familiarity, and perceived control, rather than objective probability alone. Slovic's research demonstrated that radiation-related risks are often perceived as highly uncertain and potentially catastrophic, influencing stress responses and decision-making in high-risk environments. In alignment with these findings, the RSSQ was designed to capture the psychological dimensions of radiation-related stress and the related risk perception.

Analysis

Descriptive Statistics and Preliminary Analyses:

Means, standard deviations, were calculated for all variables in this study. Demographic data was analyzed using Chi-squared tests to document potential skewed distributions toward the RW or IVR groups.

Confirmatory Factor Analysis

Confirmatory Factor Analysis (CFA) was conducted on the seven elements in the RSSQ to identify converging factors.

Independent Samples Comparisons:

Independent samples t-tests were performed to compare factors identified through the CFA between the IVR and RW conditions, thereby testing stress equivalence in the conditions. Additionally, t-tests were conducted to compare task completion times between the two conditions, to account for potential elevated cognitive load due to prolonged experimental duration in one of the conditions.

Regression and Moderation Analyses:

Multiple linear regression analyses were conducted to examine the extent that perceived stress, risk taking propensity, and the type of group were significant predictors of the radiation-induced stress factors identified through the CFA.

Presence Analysis:

Pearson correlation coefficients were calculated to explore the level correlations between Presence and its elements and the radiation-induced stress factors identified through the CFA.

CHAPTER 4. RESULTS

Before the experiment began, participants completed surveys gathering data about their demographics, as well as data assessing their risk-taking propensity and perceived stress. The following sections present the demographic, risk taking propensity, and perceived stress data and comparisons between the RW and IVR groups.

Demographics

Age

	18-24	25-34	35-44	45-54	55-65
RW	6	10	5	3	2
IVR	4	13	7	4	1
TOTAL	10	23	12	7	3

Table 1: Age distribution

Table 1 displays the distribution of participants by age group across the RW and IVR groups. A Chi-squared test of independence was conducted to determine whether age distribution differed significantly between the two groups. The test revealed no evidence for statistically significant differences in age distribution between the groups, $\chi^2(4, N = 55) = 1.44, p = .837$.

Gender

	Male	Female
RW	22	4
IVR	27	2
TOTAL	49	6

Table 2: Gender distribution

Table 2 presents the distribution of participants by gender across the RW and IVR groups. A Chi-squared test of independence was conducted to assess whether gender distribution differed significantly between the two groups. The test revealed no evidence for statistically significant difference between group assignment and gender, $\chi^2(1, N = 55) = 0.33, p = .565$.

Years of experience

	<1 year	years	2-5 years	5-10 years	10+ years
RW	5	5	3	7	6
IVR	13	6	2	5	3
TOTAL	18	11	5	12	9

Table 3: Years of experience distribution

Table 3 displays the distribution of participants according to their years of experience working with radiation or radiation related equipment. A Chi-squared test of independence was performed to evaluate whether experience levels differed significantly between the RW and IVR groups. The test indicated no evidence for statistically significant difference in years of experience between the groups, $\chi^2(4, N = 55) = 5.03, p = .284$.

Responders' competencies

The National Fire Protection Association 470 (NFPA, 2022) establishes standardized competency levels for personnel involved in hazardous materials/weapons of mass destruction (HazMat/WMD) response. These levels are Awareness, Operations, Technician, Officer, and Incident Commander, define progressive tiers of knowledge, skills, and responsibilities, as follows:

- At the Awareness level, individuals are expected to recognize potential HazMat/WMD incidents, initiate protective actions, and notify appropriate authorities.
- Operations-level responders take defensive actions to protect people, the environment, and property, often working under guidance without direct contact with the substance.
- Technicians operate at the highest tactical level, performing offensive strategies such as leak control, plugging, patching, and decontamination.

- Officers (also referred to as Hazardous Materials Officers) are responsible for supervising HazMat teams, making tactical decisions, and ensuring proper implementation of safety measures.
- The Incident Commander holds ultimate responsibility for managing the overall incident, coordinating multi-agency resources, establishing response objectives, and maintaining scene safety.

Each NFPA 470 competency level also includes foundational radiation training and experience tailored to the expected role and responsibilities of the responder.

- At the Awareness level, personnel receive basic instruction on identifying radiation hazards, understanding common warning signs (such as placards and labels), and recognizing the potential risks associated with radioactive materials.
- Operations-level responders build on this knowledge with training in radiological protective actions, use of personal protective equipment (PPE), and basic detection instruments to support life-saving and defensive operations in radiological environments.
- Technicians receive more advanced training in radiation detection and measurement, contamination control, decontamination procedures, and interpretation of instrument readings to support offensive operations involving radioactive materials.
- HazMat Officers are expected to have sufficient radiation response training to evaluate operational plans, ensure team safety, and make informed tactical decisions involving radiological threats.
- Finally, the Incident Commander requires a broad understanding of radiological concepts, risk communication, regulatory considerations, and integration of radiological subject matter experts to effectively manage complex incidents involving radioactive materials.

Training at each level is designed to align with the scope of expected duties, ensuring that responders operate safely and effectively in radiological response scenarios.

None of the participants were at the competency level of ‘None’ or ‘Awareness’. The Incident Commander category included the HazMat Officer due to the parallel in managerial duties. Table 4 presents the distribution of participants based on their competency level.

	None	Awareness	Operations level	HazMat Technician	Incident commander qualification
RW	0	0	16	9	1
VR	0	0	19	10	0
TOTAL	0	0	35	19	1

Table 4: Types of certification distribution

A Chi-squared test of independence was conducted to assess whether distribution of competency level differed significantly between the groups. The test revealed no evidence for statistically significant difference of certification distribution between the groups, $\chi^2 (2, N = 55) = 1.15, p = .563$.

Frequency of Interaction with Radiation Equipment

Participant interaction with radiation-related equipment is a dimension that needs to be accounted for due to the experimental design requiring participants to use such equipment in the experiments.

	Weekly	Monthly	Never	No rad Equipment Available
RW	2	10	10	4
VR	0	16	7	6
TOTAL	2	26	17	10

Table 5: Frequency of interaction with radiation equipment distribution

Table 5 presents the distribution of participants based on their frequency of interaction with radiation-related equipment. A Chi-squared test of independence was conducted to determine whether frequency distribution was statistically significantly different between the groups. The test revealed no evidence for statistically significant difference, $\chi^2 (3, N = 55) = 4.16, p = .244$.

Self-Reported Understanding Risks with Radiation Exposure

Participants were asked to report their perception of level of understanding of the risk associated with exposure to radiation.

	Completely	Kinda	Neutral	Kinda not	Not at all
RW	3	16	5	2	0
VR	7	8	10	1	3
TOTAL	10	24	15	3	3

Table 6: Understanding of risks associated with radiation exposure distribution

Table 6 presents the distribution of participants' self-reported understanding of the risks associated with radiation exposure. To examine whether the distribution of responses was statistically significantly different between the groups, a Chi-squared test of independence was conducted. The test revealed no evidence for statistically significant difference between the groups, $\chi^2 (4, N = 55) = 9.13, p = .0586$.

Risk-taking Propensity

The Risk-taking propensity survey collected data on participants' tendency to take risks. Table 7 displays the means and standard deviation of risk-taking propensity for the groups.

Group	n	Mean	Std dev
VR	29	3.047	0.7330
RW	26	3.091	0.6486

Table 7: Risk-taking Propensity Scores Distribution

A two-tailed student T-test was conducted to compare the risk-taking propensity scores. No evidence for a statistically significant difference in risk propensity main effect was detected, $t(53) = 0.23422, p = .8157$.

Perceived Stress

Table 8 shows the distribution of Perceived Stress by group.

Group	N	Mean	Std dev
VR	29	1.97	0.6113
RW	26	2.219	0.4472

Table 8: Perceived Stress Scores Distribution

The two-tailed student t-test was conducted to compare perceived life-stress between the groups. No evidence for a statistically significant different main effect was detected in perceived stress, $t(53) = 1.7075, p = .0936$.

Radiation-induced stress

The overarching research question was: To what extent radiation-induces stress is comparable between the RW experimental group and the IVR experimental group.

Post experiments, participants filled out a survey that included a radiation induces stress questionnaire. Table 9 presents the means and standard deviations for each group on each question:

	Group	Mean	StDev.
Q1. How much were you thinking of the radiation source during the simulation?	RW	5.423	1.6291
	IVR	5.586	1.8807
Q2. How dangerous did you perceive the radiation source to be during the exercise?	RW	2.731	1.3873
	IVR	4.466	1.5055
Q3. How concerned were you about being exposed to radiation during the exercise?	RW	2.654	1.6478
	IVR	3.621	2.1780
Q4. How comfortable did you feel getting close to the radiation source?	RW	2.577	1.6043
	IVR	3.621	1.9895
	RW	2.731	1.6866

Q5. How much did the proximity to the radiation source contribute to your overall sense of stress or anxiety?	IVR	2.483	1.7652
Q6. How cautious or hesitant were you when approaching the radiation source?	RW	2.692	1.5432
	IVR	3.103	1.7391
Q7. Did the presence of the radiation source affect your focus and concentration during the exercise?	RW	2.269	1.4016
	IVR	2.345	1.5874

Table 9. Radiation-induced stress means and standard deviations

Confirmatory Factor Analysis

To examine the latent structure underlying participants' responses to questions measuring risk-induced stress in a simulated radiation exposure scenario, a confirmatory factor analysis (CFA) was conducted using maximum likelihood extraction with Varimax rotation. The analysis included the seven items in Table 9 above.

Sampling Adequacy and Model Fit

The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was .777, indicating that the sample was meritorious (Kaiser, 1974). Bartlett's test of sphericity was statistically significant, $\chi^2(21) = 150.35$, $p < .001$, supporting the factorability of the correlation matrix.

Factor Extraction and Loadings

An initial examination of the eigenvalues suggested that two factors should be retained. The eigenvalues for the first three components were 2.70, 1.37, and 0.89, respectively. According to the Kaiser criterion (eigenvalues > 1), the first two factors were retained, accounting for a cumulative 43.29% of the variance in the data (23.38% by Factor 1 and 19.90% by Factor 2). The scree plot (Figure 8) supported this decision, displaying a clear inflection after the second component.

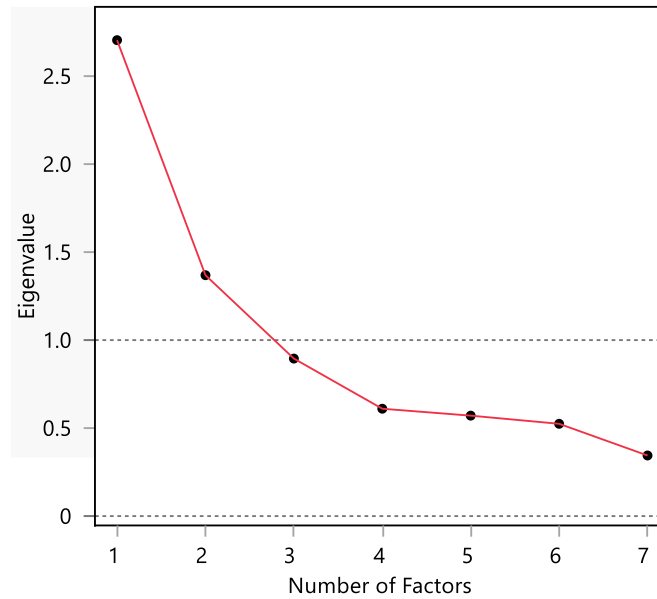


Figure 8: Radiation-induced stress Scree Plot

The significance test for the overall model yielded a Chi-square of $\chi^2(21) = 80.79$, $p < .0001$, indicating that at least one common factor exists. A subsequent test evaluating the adequacy of a two-factor solution yielded a non-significant result, $\chi^2(8) = 4.86$, $p = .773$, suggesting that two factors adequately explained the covariance among the items.

Measures of Model Fit

Model fit indices supported the adequacy of the two-factor model. The Tucker-Lewis Index (TLI) was 1.138, and the Root Mean Square Error of Approximation (RMSEA) (Browne, M. W., & Cudeck, R., 1993) was 0.00001 indicating excellent model fit. The AIC and BIC values (-10.70 and -26.76, respectively) further supported the adequacy of the two-factor solution, indicating a good balance between model fit and parsimony.

Factor Structure and Interpretation

Rotated factor loadings (with a suppression threshold of 0.30) indicated a clear pattern of item clustering. The loadings are summarized in Table 10 below:

Item	Factor 1	Factor 2
Q7 – Radiation affecting concentration	0.85	—
Q6 – Hesitation approaching source	0.63	—
Q5 – Stress from proximity	0.61	—
Q3 – Concern about exposure	—	0.70
Q2 – Perceived danger of radiation	—	0.58
Q4 – Comfort near the source	0.34	0.57
Q1 – Thinking of the source	—	0.34

Table 10: Factor Loadings

The rotated factor solution showed the following groupings:

- Factor 1, labeled ***Behavioral and Cognitive Response to Presence of Radiation Source***, included high loadings from:
 - Q7: “Did the presence of the radiation source affect your focus and concentration during the exercise?”
 - Q6: “How cautious or hesitant were you when approaching the radiation source?”
 - Q5: “How much did the proximity to the radiation source contribute to your overall sense of stress or anxiety?”
 - Q4: “How comfortable did you feel getting close to the radiation source?”

Factor 2, labeled ***Perceived Risk and Emotional Concern***, included high loadings from:

- Q1: “How much were you thinking of the radiation source during the simulation?”
- Q2: “How dangerous did you perceive the radiation source to be during the exercise?”
- Q3: “How concerned were you about being exposed to radiation during the exercise?”
- Q4: “How comfortable did you feel getting close to the radiation source?”

Q4 loaded moderately on both factors, suggesting it captures elements of both the cognitive and behavioral responses. However, the strong loading of Q7 and Q6 on Factor 1, and of Q3 and Q2 on Factor 2, indicates strong support for the conceptual clarity between the two factors.

Assessment of assumptions for parametric analysis:

Normality was assessed using the Shapiro-Wilk test for each factor within both the RW and IVR groups. Results indicated no evidence for significant departures from normality:

- **Factor 1:** RW group ($p = .185$), IVR group ($p = .264$)
- **Factor 2:** RW group ($p = .436$), IVR group ($p = .782$)

Equality of variances was evaluated using Levene's Test:

- **Factor 1:** $p = .679$
- **Factor 2:** $p = .184$

The results of the assessments above confirm that the assumptions of normality and homogeneity of variances were met. Therefore, parametric t-tests are appropriate for comparing group means for both factors. Table 11 provides means and standard deviations for each factor by group:

	Group	N	Mean	Std dev
Factor 1	IVR	29	2.89	1.31
	RW	26	2.57	1.20
Factor 2	IVR	29	4.32	1.29
	RW	26	3.35	1.06

Table 11: Means and standard deviations for the groups for both factors.

Now once the factors have been revealed the hypotheses for the research question can be presented as follows:

H_0^1 : There is no significant difference between the groups in the scores for Factor 1,
Behavioral and Cognitive Response to the Presence of a Radiation Source.

H_0^2 : There is no significant difference between the groups in the scores for Factor 2,
Perceived Risk and Emotional Concern.

The alternative hypothesis is as follows:

H_A^1 : There is a significant difference between the groups in the scores for Factor 1,
Behavioral and Cognitive Response to the Presence of a Radiation Source.

H_A^2 : There is a significant difference between the groups in the scores for Factor 2,
Perceived Risk and Emotional Concern.

To evaluate hypotheses H^1 and H^2 , independent samples t-tests were conducted.

The comparison of Factor 1 scores between the RW group ($M = 2.57$) and the IVR group ($M = 2.89$) yielded a non-significant result, $t(53) = 0.94$, $p = .350$. Thus, based on the evidence, hypothesis H_0^1 cannot be rejected.

For Factor 2, the RW group had a mean score of ($M = 3.35$), while the IVR group scored higher with a mean of ($M = 4.32$). The difference was statistically significant, $t(53) = 2.01$, $p = .0035$. The calculated Cohen's d was 0.82, pointing to a large effect size. Thus, based on the evidence, hypothesis H_0^2 is rejected and hypothesis H_A^2 is accepted.

Multiple Linear Regression Analyses

Although several demographic and contextual variables were collected, including gender, years of experience, responder's official competencies, frequency of interaction with radiation equipment, and self-reported understanding of risks associated with exposure, none of these variables demonstrated statistically significant differences between the RW and IVR groups in the preliminary group comparison analyses above.

Given the limited sample size ($N = 55$), including all available variables in a multiple linear regression would have increased the risk of model overfitting and reduced statistical power. Therefore, to ensure an interpretable model, only the variables that (a) showed variation across participants and (b) with potential relevance to the outcome based on the theoretical or empirical considerations were included. Specifically, perceived stress, risk-taking propensity, and group type were selected as predictors in the regression models.

Factor 1: Behavioral and Cognitive Response to Presence of Radiation Source

The following hypothesis is presented to assess the extent to which type of group, risk-taking propensity, and perceived stress predict scores on factors 1.

H_0^3 : None of the variables of type of group, risk-taking propensity, and perceived stress are significant predictors of the scores of factor 1.

The alternative hypothesis is as follows:

H_A^3 : At least one of the variables of type of group, risk-taking propensity, and perceived stress is a significant predictor of the scores of factor 1.

Multiple linear regression analysis was conducted to test hypothesis H_0^3 . The overall regression model for factor 1 was statistically significant, $F(3, 51) = 4.58$, $p = .0065$, indicating that the

combination of predictors explained a significant portion of the variance in factor 1 scores. The model accounted for approximately 21.2% of the variance, $R^2 = 0.212$, with an adjusted R^2 of 0.166.

Individual Predictors

- Evidence showed that **Risk-Taking Propensity** was a significant negative predictor of Factor 1 ($B = -0.665$, $t = -2.93$, $p = .0051$), suggesting that individuals with higher risk-taking tendencies tended to score lower on *Behavioral and Cognitive Response to Presence of Radiation Source*.
- Evidence showed that **Perceived Stress** was not a significant predictor ($B = 0.553$, $t = 1.89$, $p = .0643$).
- Evidence showed that **Group Type (RW vs. IVR)** was not a significant predictor ($B = -0.215$, $t = -1.35$, $p = .1842$).

Model Fit and Diagnostics

The root mean square error (RMSE) was 1.15; The lack-of-fit test was not significant ($p = .856$), indicating that the model adequately captured the relationship between the predictors and Factor 1, with no meaningful patterns left unexplained. Thus, based on the evidence, hypothesis H_0^3 is rejected and Hypothesis H_A^3 is accepted.

Factor 2: Perceived Risk and Emotional Concern

The following hypothesis is presented to assess the extent to which type of group, risk-taking propensity, and perceived stress predict scores on factor 2:

H_0^4 : None of the variables of type of group, risk-taking propensity, and perceived stress are significant predictors of factor 2 scores.

The alternative hypothesis is as follows:

H_A^4 : At least one of the variables of type of group, risk-taking propensity, and perceived stress is a significant predictor of the scores of factor 2.

Multiple linear regression analysis was conducted to test hypothesis **H_0^4** . The regression model for factor 2 was statistically significant, $F(3, 51) = 4.91, p = .0045$, indicating that the set of predictors explained a significant portion of the variance in factor 2 scores. The model accounted for 22.4% of the variance, $R^2 = 0.224$ and an adjusted R^2 of 0.178.

Individual Predictor

- Evidence showed that **Group Type (RW vs. IVR)** was a significant negative predictor of factor 2 ($B = -0.496, t = -3.10, p = .0032$), indicating that participants in the IVR group had significantly higher ***Perceived Risk and Emotional Concern*** scores than those in the RW group.
- Evidence showed that **Risk-Taking Propensity** was a significant negative predictor ($B = -0.486, t = -2.13, p = .0382$), suggesting that individuals with lower risk-taking tendencies reported higher Perceived Risk and Emotional Concern scores.
- Evidence showed that **Perceived Stress** was not a significant predictor ($B = 0.149, t = 0.51, p = .6148$), indicating no clear association between perceived stress and factor 2 scores in this model.

Model Fit and Diagnostics

The root mean square error (RMSE) was 1.154, and the lack-of-fit test was not significant ($p = .3517$), indicating that the model provides a good fit to the data without notable lack of fit.

Thus, based on the evidence, hypothesis H_0^4 is rejected, and hypothesis H_A^4 is accepted.

Time Completion

While task completion time is not a primary variable in this study, it is reported in Table 12 to account for potential differences in cognitive load between the groups. A two-tailed Student's *t*-test comparing completion times revealed a statistically significant difference, $t(53) = -2.10$, $p = .0416$, indicating that the mean completion time for the IVR group was significantly higher than that of the RW group. However, given the short duration of the task (4–5 minutes), it is reasonable to assume that time-induced cognitive load was not meaningfully elevated in either group.

Group	N	Mean	Std dev
IVR	29	287	75
RW	26	247	66

Table 12: Time to Completion in Seconds

Presence in VR Simulation

The Pearson correlation analysis was conducted to examine the relationships between two factors (factor 1 and factor 2) and six key elements characterizing the IVR simulation experience: Presence, Control, Sensory, Realism, Distraction, and Involvement. Table results indicate the strength and direction of linear relationships, along with corresponding *p*-values assessing statistical significance. Table 13 presents the descriptive statistics and Pearson product-moment correlation coefficients for each of the presence-related variables in relation to factors 1 and 2.

	Mean	Std dev	Factor	<i>r</i>	<i>p</i> -value
Presence Total	5.14	0.89	1	0.18	.3489
			2	0.26	.1709
Control Factors	5.14	1.04	1	.12	.5236
			2	0.16	.4191
Sensory Factors	5.61	0.89	1	0.16	.4056
			2	0.39	.0375*
Realism Factors	5.10	0.98	1	0.18	.3622
			2	0.25	.1913

Distraction Factors	4.36	1.13	1	0.17	.3864
			2	0.15	.4333
Involvement Factors	5.184	0.98	1	0.16	.4165
			2	0.30	.1128

Table 13: Presence Scores

Overall, correlation coefficients (r) across both factors and IVR characteristics ranged from .12 to .39, indicating weak to moderate associations. Among these, only one statistically significant relationship emerged: between Factor 2 and Sensory Factors ($r = .39$, $p = .0375$), suggesting that greater intensity or contribution of Factor 2 is moderately associated with heightened sensory engagement in the IVR environment. This finding meets the conventional threshold of statistical significance ($p < .05$), as denoted in the table.

Presence Total was modestly correlated with both factors ($r = .18$ for factor 1 and $r = .26$ for factor 2), though neither correlation reached significance ($p = .3489$ and $p = .1709$, respectively). Similarly, Control Factors demonstrated weak correlations with factor 1 ($r = .12$) and factor 2 ($r = .16$), but without non-significance.

For Realism Factors, small correlations were noted for both factor 1 ($r = .18$, $p = .3622$) and factor 2 ($r = .25$, $p = .1913$), again not meeting the threshold for significance. Distraction Factors and Involvement Factors followed a comparable pattern, showing weak relationships (r values ranging from .15 to .30) and non-significance across both factors.

In summary, the evidence supports the emergence of only one meaningful correlation, specifically between Factor 2 and Sensory Factors, suggesting that this component of the IVR simulation is particularly influenced by characteristics captured by Factor 2. All other relationships were statistically non-significant, indicating that the factors had limited influence on the remaining dimensions of the IVR experience in this sample.

CHAPTER 5. DISCUSSION

The present study significantly contributes to the ongoing discourse surrounding the application and efficacy of IVR in hazardous materials training by comparing stress responses elicited in IVR and RW simulations of radiological incidents among experienced first responders. This comparative analysis provides important insights into how different training environments influence stress reactions, training effectiveness, and operational readiness.

Interpretation of Core Findings

Both factors were measured using a 7-point Likert scale, where 1 indicated 'no stress' and 7 indicated 'an extreme level of stress.' The confirmatory factor analysis revealed two central dimensions of stress: Behavioral and Cognitive Response to the Presence of Radiation (Factor 1) and Perceived Risk and Emotional Concern (Factor 2).

The analysis showed no statistically significant difference between the IVR and RW groups for Factor 1, suggesting that both simulation modalities elicited comparable levels of cognitive vigilance and behavioral caution—critical components for maintaining operational safety during hazardous incidents. The mean scores for ***Behavioral and Cognitive Response*** were 2.57 for the RW group and 2.89 for the IVR group. These scores, both falling well below the midpoint of the 7-point Likert scale, reflect generally low levels of behavioral and cognitive stress response across both groups. This finding indicates that although participants remained cognitively and behaviorally aware of the potential hazards, the overall level of stress experienced in these domains was modest.

In contrast, a statistically significant difference was observed for Factor 2, ***Perceived Risk and Emotional Concern***. IVR participants reported higher levels of perceived risk and emotional concern (mean = 4.32) compared to their RW counterparts (mean = 3.35). The RW group's mean

score, slightly below the midpoint of the scale, reflects a low-to-moderate level of perceived risk and emotional concern. Meanwhile, the IVR group's mean score, slightly above the midpoint, suggests a moderate level of emotional engagement and perceived threat. This pattern potentially indicates that the immersive nature of the IVR environment effectively heightened emotional and risk-related responses, thereby enhancing participants' cognitive and emotional processing related to the radiation hazard. However, while the elevated stress response in IVR may benefit training by increasing emotional salience and risk awareness, there is a potential concern that the excessive emotional stimulation could lead to an inflated perception of threat. This inflation might distort realistic threat appraisals, which in turn can undermine decision-making accuracy and performance during actual emergency situations.

These findings align with and expand upon Lazarus and Folkman's (1984) cognitive appraisal theory of stress, emphasizing that stress reactions are influenced by individual threat evaluations and available coping resources. IVR likely heightens subjective risk appraisals by providing fewer implicit safety cues compared to RW scenarios, where visible expert oversight may reduce perceived risk.

The significant negative correlation found between risk-taking propensity and stress responses highlights the moderating role of individual psychological traits. Individuals with higher risk-taking tendencies consistently reported lower stress responses, supporting existing psychological literature on personality factors as critical moderators of stress resilience in high-risk environments.

Practical implications underscore IVR's potential to significantly enhance hazardous materials training, particularly where RW training opportunities are constrained. IVR offers safe, repeated, and controlled exposure to various hazardous scenarios, improving readiness and

psychological preparedness. Nevertheless, IVR-induced emotional intensity necessitates careful scenario design incorporating realistic safety cues, expert guidance, and simulated peer interactions to balance emotional responses and ensure accurate risk assessments.

Additionally, increased task completion times and cognitive loads in IVR highlight practical considerations in simulation design. IVR participants experienced greater cognitive effort interacting with virtual interfaces. Thus, future IVR training should emphasize intuitive and user-friendly designs to minimize unnecessary cognitive burdens and maximize training effectiveness and user acceptance.

Broader Context and Theoretical Implications

This study's findings extend previous research in hazardous IVR scenarios. Bayouth & Keren (2019) explored situational awareness among firefighters in hazardous IVR environments, while Engelbrecht et al. (2019) acknowledged IVR's scalability and cost-effectiveness alongside limitations related to skill transfer consensus and training outcome quality. Narciso et al. (2023) reported lower stress responses in virtual versus real environments, and Kleygrewe et al. (2023) found comparable psychological stress responses between virtual and RW police training. The current study, however, revealed higher stress responses in IVR for perceived risk and emotional concern, suggesting variations attributable to differences in simulation fidelity, safety assurances in RW training, or participant characteristics. Further research into these factors could optimize IVR training efficacy.

Practical and Training Implications

The broader implications for emergency training derived from this study are substantial. IVR simulations clearly possess unique capabilities that traditional RW simulations might lack, particularly regarding emotional engagement and perceived realism. This heightened emotional

realism, when properly calibrated, may enhance training effectiveness by facilitating deeper emotional and cognitive processing, thus potentially improving long-term retention of skills and operational readiness.

However, the potential for IVR-induced emotional intensity to foster unrealistic risk perceptions necessitates careful scenario design and moderation. Incorporating realistic safety cues, expert guidance, or simulated peer interactions within IVR scenarios may help balance the heightened emotional responses, ensuring that trainees develop accurate perceptions of risk reflective of actual emergency conditions.

Further, the increased task completion time observed in the IVR condition points to a practical consideration of cognitive load management. While IVR can closely replicate RW sensory and spatial dynamics, it often demands additional cognitive effort to interact with virtual interfaces. Training scenarios should thus strive for intuitive and user-friendly designs, minimizing unnecessary cognitive burdens to ensure training effectiveness and user acceptance.

Broader Implications for Emergency Preparedness

This study contributes to a broader understanding of how emerging technologies such as IVR can revolutionize hazardous materials training. The findings imply that IVR can effectively complement traditional methods, particularly when logistical constraints or safety considerations limit RW training opportunities. Emergency services could integrate IVR into broader curricula to provide repeated, safe, and controlled exposures to a wide range of hazardous scenarios, thereby enhancing readiness and psychological preparedness across various threat types.

Moreover, the evidence presented encourages emergency management authorities to view IVR not merely as an alternative but as an essential augmentation to existing training

frameworks. By strategically deploying IVR alongside traditional training, organizations can significantly enhance their capability to prepare responders for rare but high-impact incidents, ultimately contributing to greater resilience and operational efficacy.

Future Research Directions

Future studies should incorporate physiological stress markers such as heart rate variability or cortisol levels alongside subjective stress measures. Longitudinal assessments of skill retention and operational performance post-training would provide insights into long-term IVR efficacy. Investigating various hazards, including chemical, biological, or explosive threats, may further clarify IVR's training strengths and limitations compared to traditional methods.

In summary, this research underscores IVR's significant potential in enhancing hazardous materials training through heightened emotional realism and cognitive engagement. While effectively replicating behavioral and cognitive responses observed in RW, IVR's unique emotional intensity must be managed thoughtfully. Addressing these challenges through careful scenario integration with traditional methods can substantially enhance emergency preparedness, responder resilience, and operational effectiveness in hazardous situations.

CHAPTER 6. GENERAL CONCLUSION

This study provides an important contribution to our understanding of how the different IVR and RW simulation environments impact stress responses in the context of radiation hazard training. The investigation revealed that while both simulations elicited stress responses, distinct differences emerged in how participants perceived risk and experienced emotional concern. Specifically, although pre-experiment measures did not differ significantly between groups, individuals in the IVR environment reported higher perceived risk and stronger emotional responses compared to their RW counterparts. This disparity could be mediated by the immersive qualities of IVR, suggesting that enhanced sensory stimulation can intensify the stress response even when the behavioral and cognitive dimensions remain largely comparable across conditions. Alternatively, RW could have reduced perceived risk and weaker emotional responses which could be explained by assurances given by the presence of the RSE and the researcher.

The findings bring to focus the influential roles of individual risk-taking propensity and perceived stress. Participants with higher risk-taking tendencies consistently reported lower stress levels across both simulation modalities, whereas elevated perceived stress correlated with stronger behavioral and cognitive responses in the face of radiation exposure. These results indicate that personal traits modulate the perception of and reaction to simulated hazards, reinforcing the need to consider individual differences in the design and implementation of hazard training programs.

Moreover, the study highlights some of the practical challenges associated with IVR, such as longer task completion times, which, while not estimated to be a factor herein due to the short simulation duration, may reflect the additional effort required to acclimate to virtual

interfaces compared to naturalistic RW interactions. Such nuances points out that while IVR can potentially offer enhanced emotional engagement and memory retention, careful calibration is essential to ensure that the induced stress is both representative of actual hazard exposure and conducive to effective training outcomes.

Considering these insights, future research should adopt longitudinal designs incorporating objective physiological measures, such as heart rate variability, to more comprehensively evaluate stress responses over time. Furthermore, expanding the scope to include other types of hazards—such as chemical spills, toxic gases, or biological threats—could enrich our understanding of how different threat profiles interact with simulation modality to influence trainee response. Ultimately, these avenues for further inquiry will be critical in optimizing hazard training protocols, ensuring that simulation fidelity and individual differences are adequately addressed to better prepare personnel for real-life hazardous environments.

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APPENDIX A. ELIGIBILITY SURVEY

Section 1: General Eligibility

- 1. Are you a professional emergency responder?**
- 2. Are you between the ages of 18 and 65?**
- 3. Have you been trained in the usage of radiation detection equipment?**

Section 2: Exclusion Criteria for Both Tracks

- 4. Are you currently pregnant?**
- 5. Do you have a pacemaker, defibrillator, or other implanted medical device?**
- 6. Are you currently experiencing any of the following conditions?**
 - Tiredness or exhaustion
 - Under the influence of alcohol or drugs
 - Hung over
 - Digestive problems
 - Emotional stress or anxiety
 - Cold, flu, headaches, migraines, or earaches

Section 3: Exclusion Criteria for VR Track Only

- 7. Have you ever experienced vertigo or simulation sickness?**
- 8. Are you highly sensitive to motion sickness?**
- 9. Do you have a history of seizures or seizure disorders?**
- 10. Are you sensitive to bright lights?**

11. Do you have any physical conditions that limit the ability to use both arms and legs (e.g., physical disability, use of wheelchair, walker, cane, etc.)?

12. Do you have pre-existing binocular vision abnormalities?

13. Do you have any of the following conditions?

- Psychiatric disorders
- Heart conditions
- Serious medical conditions
- Hearing aids

APPENDIX B. BACKGROUND SURVEYS**DEMOGRAPHIC QUESTIONS**

- I. What is your age?
 - a. 18-24 years old
 - b. 25-34 years old
 - c. 35-44 years old
 - d. 45-54 years old
 - e. 55-65 years old

- II. What is your gender?
 - a. Male
 - b. Female
 - c. Non-binary/Third gender
 - d. Prefer not to say

- III. What is your ethnicity?
 - a. White/Caucasian
 - b. Black/African American
 - c. Hispanic/Latino
 - d. Asian/Pacific Islander
 - e. Native American
 - f. Other (Please specify): _____

- IV. What is your Marital Status?
 - a. Single
 - b. Married
 - c. Divorced
 - d. Widowed
 - e. Other (Please specify): _____

- V. How many Dependents do you have?
 - a. 0
 - b. 1
 - c. 2
 - d. 3
 - e. 4
 - f. 5+

- VI. What is your level of education?
 - a. High School Diploma/GED
 - b. Some College
 - c. Associate's degree
 - d. Bachelor's Degree
 - e. Master's Degree
 - f. Doctorate/PhD

g. Other (Please specify): _____

VII. How many years of radiation related training and/or experience do you have?

- a. Less than 1 year
- b. 1-2 years
- c. 2-5 years
- d. 5-10 years
- e. More than 10 years

VIII. What is your employment type?

- a. Full-time career
- b. Part-time career
- c. Volunteer

RADIATION BACKGROUND

I. What level of certification/training in HAZMAT?

- 1) None
- 2) Awareness-level
- 3) Operations-level
- 4) Technician-level
- 5) Incident Commander-level

II. When did you achieve this certification?

- 1) Date (Year at a minimum)

III. Date of last radiation-related training

- 1) MM/YY

IV. Date of last radiation-related exercise

- 1) MM/YY

V. How frequently do you interact with radiation equipment in your routine daily work.

Use Likert scale:

- 1) Daily
- 2) Weekly
- 3) Monthly
- 4) Never
- 5) I don't have radiation equipment available to me

VI. How likely do you think it could be that a radiological incident might occur in your AOR?

- 1) Extremely likely
- 2) Likely
- 3) Possible
- 4) Unlikely

- 5) Extremely Unlikely

VII. Would you serve as an Incident Commander for radiation incidents?

- 1) Yes
- 2) No

VIII. How well do you understand the risks associated with radiation exposure?

- 1) Completely
- 2) Kinda
- 3) Neutral
- 4) Kinda not
- 5) Not at all

Please provide what you think are the primary risks associated with radiation exposure:

IX. How confident do you feel in your own ability to perform well when responding to a radiation incident?

- 1) Completely confident
- 2) Somewhat confident
- 3) Not confident at all

X. At what radiation doses would you expect to see physical symptoms, such as nausea, vomiting, skin burns? (multi select)

- 1) 500 microrem (μ rem)
- 2) 500 millirem (mrem)
- 3) 5 rem
- 4) 10 rem
- 5) 25 rem
- 6) 50 rem
- 7) 100 rem
- 8) 400 rem

APPENDIX C. APPROVAL FOR RESEARCH (IRB)

IOWA STATE UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Institutional Review Board
Office of Research Ethics
Vice President for Research
2420 Lincoln Way, Suite 202
Ames, Iowa 50014
515 294-4566

Date:	12/04/2024	
To:	Amon McAllister	Nir Keren
From:	Office of Research Ethics	
Title:	Comparing Participant Responses to Simulated Radiation Scenarios in Virtual Reality and Real-World Environments	
IRB ID:	24-479	
Submission Type:	Initial Submission	Review Type: Full Committee
Approval Date:	12/03/2024	Approval Expiration Date: 12/03/2025

The research project referenced above has received approval from the Institutional Review Board (IRB) at Iowa State University. Please refer to the IRB ID number shown above in all correspondence regarding this study.

To ensure compliance with federal regulations (45 CFR 46 & 21 CFR 56), please be sure to:

- Use only the approved study materials, including the recruitment materials and informed consent documents that have the IRB approval stamp.
- [Retain signed informed consent documents](#) for 3 years after study closure when documented consent is required.
- Obtain IRB approval prior to implementing any changes to the study or study materials, including addition or removal of study personnel or changes to their institutional affiliation. Approval of personnel applies only to their institutional affiliation indicated in the most recent approved application.
- Inform the IRB if the Principal Investigator and/or Supervising Investigator end their role or involvement with the project with sufficient time to allow an alternate PI/Supervising Investigator to assume oversight responsibility. Projects must have an eligible PI to remain open.
- Promptly inform the IRB of any addition of or change in federal funding for this study. Approval of this project applies only to funding sources that are specifically named in the corresponding IRB application.
- Immediately inform the IRB of (1) all serious and/or unexpected [adverse experiences](#) involving risks to subjects or others; and (2) any other [unanticipated problems](#) involving risks to subjects or others.
- IRB approval means that you have met the requirements of federal regulations and ISU policies governing human subjects research. Approval from other entities may also be needed. For example, access to data from private records (e.g., student, medical, or employment records, etc.) that are protected by FERPA, HIPAA, or other confidentiality policies requires permission from the holders of

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